Project Report: Cost & Financial Analysis

1. Project Overview

This project focuses on analyzing project cost and financial performance across multiple projects. The goal is to identify budget deviations, outstanding payments, and overall project health. The analysis helps stakeholders make informed decisions by tracking key financial metrics such as budget allocation, actual costs, and payment recovery status.

2. Dataset Summary

The dataset contains project-level financial and operational details. Key fields include:
- Project ID and Project Name
- Budget (INR) and Actual Cost
- Start Date and Project Status
- Invoice Amount and Payment Received

The dataset is used to evaluate cost performance, calculate outstanding payments, and determine whether projects are exceeding their allocated budgets.

3. Exploratory and Data Analysis using python.

The data was processed and cleaned using Python in Jupyter Notebook. Below are the key steps performed:

- Original Data

	Project_ID	Project Name	Contractor	Region	Budget (INR)	Actual Cost	Invoice Amt	Payment Recvd	Outstanding	Start Date	End Date	Status
0	P001	Highway Expn L1	ABC Infra	North	5000000	5200000	3000000	2500000.0	NaN	01-01-2024	30-12-2024	On going
1	P002	Bridge Work-A	XYZ Constructions	East	7000000	6800000	4000000	4000000.0	0.0	15/02/2024	15/11/2024	Completed
2	P003	Metro Phase 2	LMN Pvt Ltd	West	12000000	12500000	8000000	6000000.0	2000000.0	2024-03-01	2025-06-30	Ongoing
3	P004	Road Repair Z5	ABC Infra	North	3000000	2800000	2000000	NaN	1000000.0	05-04-2024	30-09-2024	Delayed
4	P005	Airport Extn	XYZ Constructions	South	15000000	14000000	9000000	8500000.0	500000.0	2024/05/01	2025/01/30	On Track

- Standardized column names by removing extra spaces and replacing spaces with underscores.
- Cleaned inconsistent status values by replacing 'On going' with 'Ongoing'.
- Converted Start_Date column into proper datetime format.
- Handled missing values in Payment_Recvd by replacing them with 0.
- Calculated Outstanding amount as the difference between Invoice Amount and Payment Received.
- Computed Cost Variance as the difference between Actual Cost and Budget.
- Created an Over Budget Flag to identify projects exceeding their budget.

- Final Data

Contractor	Region	Total_Budget	Cost_Actual	Invoice_Amt	Payment_Recvd	Outstanding	Start_Date	End_Date	Status	Cost_Variance	Over_Budget_Flag
ABC Infra	North	5000000	5200000	3000000	2500000.0	500000.0	2024-01-01	2024-12-30	Ongoing	200000	Yes
XYZ onstructions	East	7000000	6800000	4000000	4000000.0	0.0	2024-02-15	2024-11-15	Completed	-200000	No
.MN Pvt Ltd	West	12000000	12500000	8000000	6000000.0	2000000.0	2024-03-01	2025-06-30	Ongoing	500000	Yes
ABC Infra	North	3000000	2800000	2000000	0.0	2000000.0	2024-04-05	2024-09-30	Delayed	-200000	No
XYZ onstructions	South	15000000	14000000	9000000	8500000.0	500000.0	2024-05-01	2025-01-30	On Track	-1000000	No

4. Data Analysis using SQL

Preformed structured analysis in SQL to answer key business question.

1. **Status of Project** –

Status	Total_Projects
Ongoing	14
Completed	7
Delayed	7
On Track	4
Over Budget	3

2. **Top 5 highest cost overrun projects**

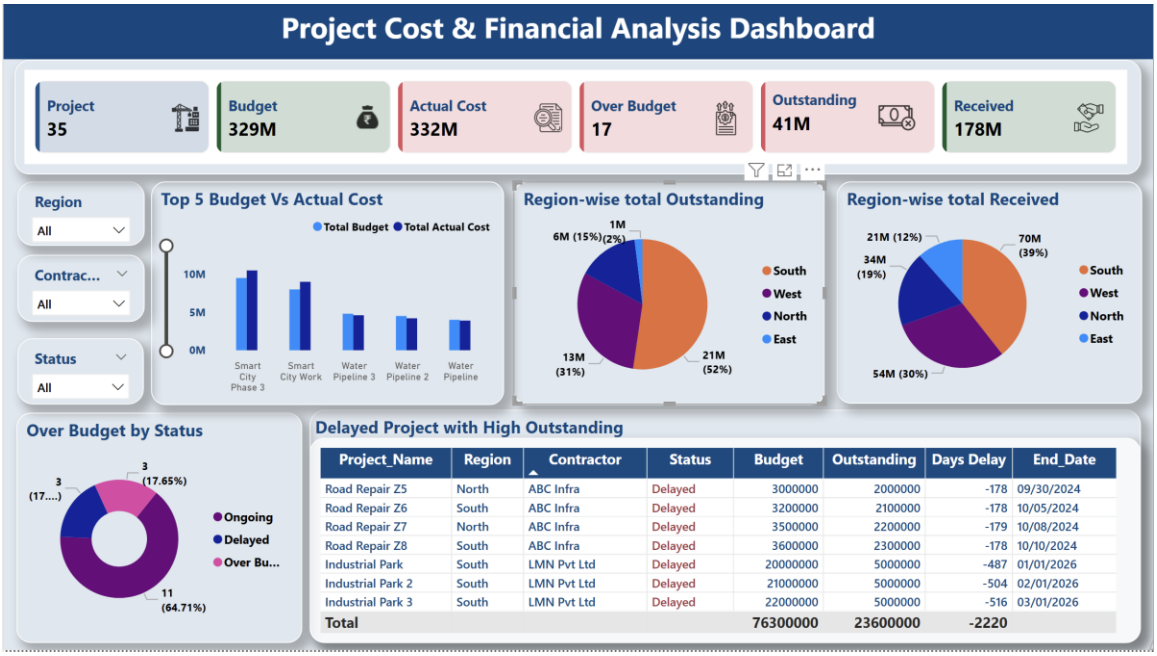
Project_ID	Project_Name	Total_Budget	Cost_Actual	Cost_Variance
P029	Industrial Park 3	22000000	23000000	1000000
P026	Smart City Phase 3	9500000	10500000	1000000
P016	Smart City Phase 2	9000000	10000000	1000000
P023	Metro Phase 4	14000000	15000000	1000000
P009	Industrial Park	20000000	21000000	1000000

3. **Calculate Delay project duration (in days)**

Project_ID	Project_Name	Days_Delay
P033	Metro Phase 5	533
P023	Metro Phase 4	516
P029	Industrial Park 3	516
P019	Industrial Park 2	504
P013	Metro Phase 3	502

5. Dashboard in Power BI

Finally create a Dashboard in Power BI to present insights visually.



Project_Name	Region	Contractor	Status	Budget	Outstanding	Days Delay	End_Date
Road Repair Z5	North	ABC Infra	Delayed	3000000	2000000	-178	09/30/2024
Road Repair Z6	South	ABC Infra	Delayed	3200000	2100000	-178	10/05/2024
Road Repair Z7	North	ABC Infra	Delayed	3500000	2200000	-179	10/08/2024
Road Repair Z8	South	ABC Infra	Delayed	3600000	2300000	-178	10/10/2024
Industrial Park	South	LMN Pvt Ltd	Delayed	20000000	5000000	-487	01/01/2026
Industrial Park 2	South	LMN Pvt Ltd	Delayed	21000000	5000000	-504	02/01/2026
Industrial Park 3	South	LMN Pvt Ltd	Delayed	22000000	5000000	-516	03/01/2026
Total				76300000	23600000	-2220	

6. Business Recommendations.

- **Implement strict cost control measures** to reduce the high number of over-budget projects.
- **Set budget variance alerts (e.g., >10%)** to take early corrective action.
- **Improve cash flow management** by reducing ₹41M outstanding payments.
- **Introduce automated payment follow-ups** to ensure timely collections.
- **Monitor project delays regularly** using a Days Delay KPI.
- **Identify root causes of delays** (contractor, approvals, or regional issues).
- **Develop contractor performance scorecards** to track efficiency and reliability.
- **Optimize resource allocation across regions** based on performance insights.
- **Use dashboards in regular management reviews** for data-driven decision-making.
- **Adopt predictive analytics (future scope)** to forecast cost overruns and delays.